DATA DICTIONARY – CUSTOMER ANALYTICS DASHBOARD

Project Overview

This dashboard analyzes customer behavior, marketing performance, channel efficiency, and segmentation using a snowflake schema data model and advanced DAX measures.

Data Model

- Fact Table: FactCustomerMetrics
- Dimensions: DimCustomer, DimDate, DImAgeGroup, DimEducation, DimMarital.
- Supporting Tables: Channel Table, Campaign Table, Product Categoy Table.

Key Capabilities

- Channel performance analysis
- Campaign effectiveness tracking
- RFM segmentation
- Customer behavior modeling
- Time intelligence (MoM trends)

1. CUSTOMER KPIs

Total Customers

Description: Total number of unique customers in the dataset
Business Meaning: Core base to measure growth, segmentation and performance
Used In: All pages

Avg Customer Income

Description: Average income of customers
Business Meaning: Helps identify purchasing power and segment targeting
Used In: Customer Profile

Avg Customer Age

Description: Average age of customers
Business Meaning: Understand demographic distribution
Used In: Customer Profile

Avg Recency

Description: Average days since last purchase
Business Meaning: Measures customer activity and recency behavior
Used In: RFM / Retention

Customers With Children

Description: Number of customers with children
Business Meaning: Enables behavioral segmentation (family vs non-family)
Used In: Customer Profile

2. REVENUE KPIs

Total Revenue

Description: Total customer spending
Business Meaning: Main business performance indicator
Used In: All pages

Avg Purchase Value

Description: Average revenue per purchase
Business Meaning: Measures transaction value quality
Used In: Revenue Analysis

Revenue Engaged Customers

Description: Revenue generated by engaged customers
Business Meaning: Identifies value contribution of active users
Used In: Engagement Analysis

Revenue by Product Category (Wines, Meat, Fish, etc.)

Description: Revenue split by product category
Business Meaning: Product performance analysis
Used In: Product Analysis

3. PURCHASE BEHAVIOR KPIs

Total Purchases

Description: Total number of purchases
Business Meaning: Volume indicator of customer activity

Avg Purchases Customer

Description: Average purchases per customer
Business Meaning: Frequency indicator

Web / Store / Catalog Purchases

Description: Purchases by channel
Business Meaning: Channel behavior understanding

Deals Purchases

Description: Purchases made through promotions
Business Meaning: Sensitivity to discounts

4. CHANNEL PERFORMANCE

Web Conversion Rate

Description: Ratio of web purchases over web visits
Business Meaning: Measures website effectiveness

Web Purchase % / Store % / Catalog % / Deals %

Description: Channel share distribution
Business Meaning: Channel performance comparison

Top Channel %

Description: Highest performing channel share
Business Meaning: Identifies dominant channel

Top Channel Name

Description: Name of the best performing channel
Business Meaning: Quick insight for decision making

Purchases by Channel

Description: Dynamic calculation of purchases by selected channel
Business Meaning: Enables flexible channel analysis

Revenue by Channel

Description: Revenue allocation by channel
Business Meaning: Channel monetization

5. MARKETING CAMPAIGN KPIs

Campaign Responses

Description: Number of customers who accepted campaigns
Business Meaning: Campaign effectiveness

Campaign Conversion Rate

Description: Ratio of responses over total customers
Business Meaning: Overall campaign performance

Total Campaign Accepted

Description: Total accepted campaigns
Business Meaning: Customer engagement with marketing

Revenue Campaign Responders

Description: Revenue generated by customers who responded
Business Meaning: ROI of campaigns

Revenue per Responder

Description: Average revenue per responding customer
Business Meaning: Campaign quality vs quantity

% Revenue from Responders

Description: Share of revenue from responders
Business Meaning: Campaign contribution to total revenue

Targeted Customers

Description: Customers exposed to campaigns
Business Meaning: Campaign reach

6. CAMPAIGN ANALYSIS (DYNAMIC)

Campaign Responses Dynamic

Description: Responses by individual campaign (Cmp1–Cmp5)
Business Meaning: Campaign-level performance

Targeted by Campaign

Description: Customers targeted per campaign
Business Meaning: Campaign coverage

Campaign Conversion Rate Dynamic

Description: Conversion per campaign
Business Meaning: Campaign efficiency comparison

Revenue by Campaign

Description: Revenue attributed per campaign
Business Meaning: Revenue distribution

Revenue per Responder Dynamic

Description: Revenue efficiency per campaign
Business Meaning: High-value campaign identification

Avg Campaign Conversion Rate Dynamic

Description: Average conversion benchmark
Business Meaning: Performance baseline

Avg Revenue per Responder Dynamic

Description: Average campaign revenue benchmark
Business Meaning: Efficiency baseline

7. RETENTION KPIs

Active Customers

Description: Customers active in last 30 days
Business Meaning: Short-term retention

At Risk Customers

Description: Customers inactive for long period
Business Meaning: Churn risk

Estimated CLV

Description: Customer lifetime value approximation
Business Meaning: Long-term value estimation

8. RFM SEGMENTATION

High Value Customers

Description: Customers in top engagement segments
Business Meaning: Core business drivers

VIP Customers

Description: Champions segment
Business Meaning: Highest value customers

Loyal Customers

Description: High engagement customers
Business Meaning: Stable revenue base

% High Value Customers (RFM)

Description: Share of high value customers
Business Meaning: Customer quality

% Medium / Low Engagement

Description: Distribution by engagement level
Business Meaning: Customer mix

Avg Recency At Risk

Description: Recency of low engagement customers
Business Meaning: Churn analysis

9. BEHAVIORAL METRICS (SCATTER / RFM ENGINE)

Frequency

Description: Total purchases per customer
Business Meaning: Purchase behavior intensity

Avg Frequency

Description: Average purchase frequency
Business Meaning: Segment comparison

Recency per Customer

Description: Average recency per customer
Business Meaning: Activity level

Avg Recency (Line)

Description: Rounded recency for visualization
Business Meaning: Used for reference lines

Revenue per Customer

Description: Average revenue per customer
Business Meaning: Customer value

10. PRODUCT ANALYSIS

Revenue Selected Category

Description: Dynamic revenue by selected category
Business Meaning: Flexible product analysis

Revenue % by Category

Description: Share of revenue per category
Business Meaning: Product mix

Category Rank

Description: Ranking of categories by revenue
Business Meaning: Top-performing products

Top Product Revenue

Description: Highest revenue category
Business Meaning: Best-selling product

Top Product Revenue %

Description: Share of top category
Business Meaning: Revenue concentration

Revenue Concentration %

Description: Revenue concentration across categories
Business Meaning: Diversification vs dependency

Active Categories

Description: Number of active categories
Business Meaning: Product portfolio breadth

11. GROWTH & TIME INTELLIGENCE

(Applies to Customers, Revenue, Campaigns, etc.)

Includes:

- Prev Month
- Variance
- Growth %

Business Meaning:
Track performance evolution over time

12. STATISTICAL METRICS

Median Income

Description: Median income value
Business Meaning: Outlier-resistant income analysis

Median Age

Description: Median age
Business Meaning: Better demographic central tendency

% High Income

Description: Customers above income threshold
Business Meaning: Premium segment detection

% Core Age (30–55)

Description: Customers in main age group
Business Meaning: Target audience